

Faycal Kilali

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Education

Université de Montréal – M.Sc. Computer Science (with Internship) – Montreal, QC 2026 – 2028

- Ranked **#3** for AI in Canada & **#9** worldwide (2025 Shanghai Ranking); home to **MILA**—one of the world's largest academic AI research centres, co-founded by Turing laureate **Yoshua Bengio**.
- Concentrating in AI/ML through **MILA**-affiliated coursework at UdeM.

Mount Allison University – B.Sc. (Hons) Computer Science and Mathematics – Sackville, NB 2022 – 2026

- Honours degree in CS & Mathematics; capstone in enterprise database systems (content-aware document search engine, PostgreSQL, MVC, CI/CD).
- **Founder & President**, AI & Robotics Society 2023–2026
 - Grew to **62 members**; recruited **3 faculty advisors** and secured hardware resources & funding for projects
 - Organized project plans and coordinated members around hands-on AI & robotics initiatives.
- **Co-Founder & VP**, Mature Students Club — led recruitment at club fairs, engaging **50+** prospective members; organized peer tutoring sessions in CS & mathematics. 2023–2025

Experience

Fullstack Software Developer, AI Object Detection Framework and Educational Tool – Sackville, NB May 2024 – Sep 2024

- Secured \$4,000 TD Ignite Grant and led development of AI-based object detection educational tool, implementing YOLOv10 model across Flask backend, Streamlit web interface, and Tkinter GUI with support for 243 languages
- Architected full-stack educational game integrating real-time computer vision, REST API design, and multi-platform deployment, achieving comprehensive documentation with API specifications and workflow diagrams

Mathematics and Computer Science Peer Tutor, Mount Allison University – Sackville, NB Mar 2023 – Sep 2024

- Tutored 50+ students across various mathematical subjects effectively, resulting in positive feedback from students

Technical Lead, Greater Dorchester Moving Forward – Dorchester, NB Jan 2023 – August 2023

- Spearheaded technology operations for a nonprofit organization by leading data collection, mapping, and software development—enhancing impact and advancing the mission
- Collaborated with other departments within the organization to integrate technical solutions relevant to the seated board

Selected Projects

Content-Aware Enterprise Document Search & Management System | Java, PostgreSQL (SQL), JDBC, PDFBox, Bash, MVC, Agile, Scrum, DevOps, Figma | GitHub 2025

- **Architected** a decoupled **MVC** system using **Dependency Injection** to orchestrate state management between the front-end and the application's backend, significantly reducing component coupling and enhancing system modularity
- **Designed** a hybrid search algorithm combining PostgreSQL's `tsvector` for optimized metadata retrieval ($O(\log n)$) with a custom Java-based regex engine for deep content analysis of binary PDF blobs, enabling granular pattern matching
- **Orchestrated** the team's **Agile/Scrum** workflow, serving as **Technical Lead** to mentor peers on Git best practices, resolve merge conflicts, and manage technical debt to ensure on-time delivery of prototypes
- **Established** comprehensive **DevOps** and documentation standards, authoring UML architectural diagrams and configuring **CI/CD** pipelines (Gradle) to standardize the build and release lifecycle for the team

Room Scanner & Voice Guidance System | Python, YOLOv10, Raspberry Pi, Bash, Computer Vision, Soldering, Embedded Engineering | GitHub 2024

- Engineered embedded room scanning system integrating computer vision with custom hardware circuits (GPIO, LED indicators, push-button controls), implementing both hardware and software debouncing techniques
- Developed real-time voice feedback system for object detection with respect to the physical location of the system, providing information/guidance for blind individuals throughout the observable area
- Created comprehensive documentation of the hardware schematics and the software components with automated installation scripts, thereby emphasizing reproducible deployment

Multi-Paradigm Wordle Puzzle Solver AI Agent | Python, Streamlit, Information Theory, AI | GitHub 2024

- Implemented four AI paradigms (Information-Theoretic, Model-Based, Goal-Based, Utility-Based), utilizing entropy calculations and information gain maximization to achieve optimal 3-5 guess performance
- Architected Streamlit visualization dashboard displaying real-time AI decision-making with time complexity of $O(W^2L)$, demonstrating advanced algorithm optimization and probabilistic reasoning

Multivariate Regression Framework | Python, PyTorch, NumPy, Machine Learning, AI | GitHub 2024

- Developed machine learning framework from mathematical foundations, implementing gradient descent optimization, L2 regularization (Ridge), and closed-form solutions using PyTorch and NumPy
 - Conducted systematic experimental analysis across 5 dimensions (noise sensitivity, training set impact, parameter complexity, regularization effects) with comprehensive visualizations demonstrating theoretical ML concepts in practice
- Dreamy Pastures (Multiplayer Board Game Engine)** | Java, UML, OOP, Multiplayer | GitHub 2024

- Collaboratively developed local multiplayer board game supporting up to 4 concurrent players with real-time state synchronization and turn-based game logic, demonstrating OOP Design Patterns, and extensive UML Documentation

RSA Encryption & Decryption System | Python, RSA, Security | GitHub 2021

- Implemented RSA cryptographic algorithm from mathematical foundations, including key generation using prime number factorization, modular exponentiation, and Euler's totient function with configurable prime digit lengths for customizable security levels
- Developed modular encryption system supporting both file and string encryption/decryption with Unicode numeral representation, integrating multiple in-house implementations of the algorithms (prime generation, modular arithmetic, extended GCD) into a cohesive cryptographic framework

Open Source Video Games | MBScript, Python, PHP | Floris Evolved | Tainted Paths | Medieval Conquests 2015-2019

- Founded and led development of three major game modifications with **314,000+ active subscribers** and **1.3+ million unique visitors** across platforms, achieving **90-95% positive ratings**
- Architected & developed approximately 100,000 lines of MBScript & Python code for game logic and modular systems
- Managed distributed teams of volunteers across multiple years, implementing complex features including custom gameplay mechanics and enhanced AI systems

Skills

Methodologies: Agile, Scrum, DevOps

Algorithmic Paradigms: Divide and Conquer, Greedy Algorithms, Dynamic Programming, Decrease and Conquer

Programming Languages: Python, Java, C, C++, BASH/Shell Scripting, MBScript, Assembly (MIPS, RISC-V), LaTeX, SQL

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Flask, Streamlit, YOLOv10, REST API, JDBC, PDFBox, UML

Tools & Platforms: Git, PostgreSQL, Figma, Unity, Linux/Unix, Windows, AWS, Azure, Raspberry Pi, Arduino

Mathematical: Vector Calculus, Multivariable Calculus, Advanced Linear Algebra, Mathematical Modeling, Probability and Statistics, Discrete Structures, Real Analysis, Mathematical Optimization, Modern Algebra, Differential Equations, General Physics

Computer Science: Software Design, Operating Systems, Advanced Data Structures & Algorithms, Database Systems, Computer Networks, Graph Theory, Algorithm Analysis, Systems Programming, Digital Electronics and Digital Signal Processing, Artificial Intelligence, Simulation and Modeling, Computer Organization & Architecture, Object-Oriented Design

Software Engineering: Object-Oriented Programming (OOP), MVC Architecture, System Design, Parallel Processing, Procedural Programming, Logic Programming, Database Programming

Artificial Intelligence & Machine Learning: representation learning, self-supervised & meta-learning, reinforcement learning, generative models (GANs, GFlowNets, diffusion models), natural language processing & large language models, probabilistic inference & Bayesian networks, AI agents (information-theoretic, model-based, goal-based, utility-based, logic-based), deep learning, computer vision, information theory, & AI applications in science & robotics

Embedded Systems & Hardware: Hardware Design, Hardware Schematics, Debouncing, Automation

Domain Expertise: Low-level programming, Game Engines, Security/Cryptography, Server Infrastructure, GIS, Digital Signal Processing, Computer Graphics, Haptics, Computer Networks, VR/AR, GPU, CPU

Languages: English (Native), Standard Arabic (Professional), Saudi Arabic (Professional), Darija (Native), French (MIFI)

Certifications: *Machine Learning with Python* (IBM), *Object Oriented Design* (UAlberta), *Microsoft Learn AI Skills* (Microsoft Corporation)

Awards & Accomplishments

- **UdeM Exemption Scholarship (\$11,004 CAD/year)** – Université de Montréal, 2026–2028
Awarded full exemption scholarship upon evaluation following admission to the M.Sc. Computer Science program, offsetting tuition surcharges for the full duration of the degree.
- **TD Ignite Grant Recipient (\$4,000)** – Mount Allison University, 2024
Awarded competitive grant for developing an AI-based educational tool featuring real-time object detection and multilingual support across 243 languages.
- **3rd Place Winner, The Pitch Competition (\$1,000)** – Mount Allison University
Secured recognition for presenting a scalable solution bridging AI and healthcare, showcasing hardware-software integration and algorithm design for medical assistance devices